

DR7: The Grouping Function Inherits the Wall

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Abstract

DR5 established that a proposition-ranking nominator N cannot distinguish a commitment D from any specific realisation r_i when $|\mathrm{realisations}(D)| > 1$; escape requires importing an external grouping function g . DR6/DR6c/DR6d/DR6e triangulated: the wall bites if and only if the realisation set is open at test time and the verifier lacks semantic reasoning about D . DR5 stated that "the correctness of g becomes the load-bearing question." DR7 formalises what that means and proves two consequences.

Theorem 1 (soundness–completeness gap for open realisation sets). If a grouping function g is constructed at design time and D admits realisations at test time that were not observable at design time, then g cannot be simultaneously sound and complete for D . Formally: for any g constructed as a function of a design-time realisation sample $S \subseteq \mathrm{realisations}(D, C)$, either g over-generalises (includes non-realizations of D in its class) or under-generalises (excludes realizations of D from its class), on some corpus C .

Theorem 2 (learned- g Spencer collapse). If g is learned from a signal that includes the ranker N 's output on the training data, then class-aware ranking under g decomposes into an equivalent proposition-ranking problem with a larger score function. Class awareness supplies no additional discriminating power in this case; the DR5 wall re-appears at the class level.

Together, the two theorems say: **class-aware nomination is not a complete escape route on its own**. It relocates the wall from proposition ranking to grouping-function construction. The wall is still there; it now lives in the question "*was g built with genuine access to D 's realisation structure?*" Sound and complete g is possible only when the realisation set is closed at design time (equivalent to DR5's condition (a) closed) or when D 's semantic form is available to construction (equivalent to DR5's condition (b) satisfied). Neither condition changes; the wall's antecedents transfer from N to g .

Practical consequence. DR5 said "class-aware nomination requires g ." DR7 says: **g requires the same conditions DR5 required of N** . The wall is not escaped by class-awareness; it is refactored. Genuine escape requires either closed realisation sets or semantic access to D somewhere in the pipeline. The class-aware framing helps by making the location of the missing property explicit – but it does not itself supply the property.

1. Setup

Reuse DR5's setup. Corpus C finite. Commitment D semantic; a proposition p realises D iff p asserts or presupposes D . Realisation set $\mathrm{realisations}(D, C)$. Proposition-ranking nominator $N: C \rightarrow \mathbb{R}$.

A **grouping function** for D on C is a map $g_D: C \rightarrow \{0, 1\}$ (equivalently a subset of C) such that $g_D(p) = 1$ iff g_D 's designer intends p to be grouped with realizations of D .

g_D is **sound** for D on C iff $g_D(p) = 1 \rightarrow p \in \mathrm{realisations}(D, C)$.

g_D is **complete** for D on C iff $p \in \mathrm{realisations}(D, C) \rightarrow g_D(p) = 1$.

The **class score** of D under (N, g_D) is any aggregation of $\{N(p) : g_D(p) = 1\}$: cardinality, sum, max, spread, or any function of the class members' N -scores.

2. Theorem 1 – soundness–completeness gap

Statement. Let g_D be constructed at design time as a function of a sample $S \subseteq C_{\{\text{design}\}}$ of design-time realisations, where $C_{\{\text{design}\}}$ is a design-time corpus and $C_{\{\text{test}\}} \supseteq C_{\{\text{design}\}}$ is the test-time corpus. If there exists $r \in \text{realisations}(D, C_{\{\text{test}\}})$ with $r \notin C_{\{\text{design}\}}$ (a genuinely novel realisation), then g_D cannot be simultaneously sound and complete for $C_{\{\text{test}\}}$.

Proof. g_D 's output on r^* is determined by g_D 's construction, which used only S . Two cases:

something outside S with the design-time realisation set. This can happen only if g_D over-generalises the surface features of S 's members. But an over-generalising g_D will also fire on some non-realizations $q \in C_{\{\text{test}\}}$ with the same over-general surface features – witnessed by any q syntactically similar to S 's members but semantically distinct from D . Such a q exists whenever the language of $C_{\{\text{test}\}}$ is rich enough to construct syntactic near-copies with different semantics; this is guaranteed for natural-language corpora and for code corpora with any nontrivial vocabulary. Then g_D is unsound: $g_D(q) = 1$ but $q \notin \text{realisations}(D, C_{\{\text{test}\}})$.

- **Case A:** $g_D(r^*) = 1$. Then g_D is committed to grouping

of D ; g_D is incomplete.

- **Case B:** $g_D(r^*) = 0$. Then g_D excludes a genuine realisation

Either case establishes the claim. $\square$

Empirical instance. DR6d exhibits Case B directly: `regex_verifier` was designed against 5 realisation labels; R6 was a novel realisation at test time; `regex_verifier`'s implicit g_D (via the score function) outputs 0 on R6. R6 is a realisation of naive-UTC; g_D misses it; g_D is incomplete. Case A is illustrated by DCR1f's `target_v4` firing on Maxwell's instant across whole medium at the 1880 placebo – the matcher over-generalised beyond its design-time realisations and included a case whose status as "genuine T1 realisation" versus "projection" is exactly the placebo-invalidity ambiguity DR5 §5 said is irreducible.

Corollary (open-enumeration incompatibility). For any D with $\text{realisations}(D, C_{\{\text{test}\}}) \not\subseteq C_{\{\text{design}\}}$, g_D 's soundness and completeness are jointly unattainable at design time.

3. Theorem 2 – learned- g Spencer collapse

Statement. Let $g_D^{\{\text{learned}\}}$ be a grouping function whose construction included, as training signal, the ranker N 's output on some training subset $T \subseteq C_{\{\text{design}\}}$ (equivalently: $g_D^{\{\text{learned}\}}$ was trained to predict "does this proposition realise D " using a target label that was itself derived from N 's score or its high-scoring behaviour on T). Then class-aware ranking under $(N, g_D^{\{\text{learned}\}})$ is equivalent to proposition-ranking under a score function N' : $C \rightarrow \mathbb{R}$ where $N'(p)$ depends only on p and on N 's output at p . In particular, class-awareness supplies no additional discriminating power beyond what N already had on T -similar propositions.

Proof sketch. $g_D^{\{\text{learned}\}}(p)$ can be written as $f(\phi(p), N(p))$ where $\phi(p)$ is any surface-feature vector of p and f is the learned classifier. Substituting into any class-aware aggregation $A(\{N(q) : g_D^{\{\text{learned}\}}(q) = 1\})$, the outer aggregation is a function of proposition-level scores, and the membership determination is itself a function of proposition-level features and N -scores. So the composition $A \circ (N, g_D^{\{\text{learned}\}})$ can be written as a function $C \rightarrow \mathbb{R}^k \rightarrow \mathbb{R}$ of per-proposition information alone. This is a proposition-ranking function of a different score N' ; DR5's original theorem applies to N' directly.

Consequence. Any learned g_D that pipes N 's output through its training does not escape DR5. It builds a more expensive N' and the wall reappears at N' 's level.

Empirical instance. DCR2a's target_v4 was implicitly a learned- g of a different flavour: it was hand-tuned by inspection of the DCR1e extractor's outputs (which are the "signal" analog to N 's output). DCR1f's held-out validation showed the resulting matcher's precision-recall wall – exactly the Spencer collapse DR7 Theorem 2 predicts.

4. What DR7 buys, jointly with DR5

DR5 said: proposition-ranking cannot solve multi-realisation verification; import g .

DR7 says: importing g requires either (i) closed realisation set at design time (equivalent to DR5's condition (a) closed) or (ii) independent access to D 's semantic structure (equivalent to DR5's condition (b) satisfied). Neither of DR7's conditions is weaker than DR5's. The wall does not disappear when class-awareness is imported; it *relocates* from N to g .

The joint claim (DR5+DR7):

Genuine multi-realisation commitment verification requires either semantic access to D^{**} .

- **the realisation set is closed at design time**, or
- **at least one component of the verification pipeline has independent**

If both conditions fail, no proposition-independent verification architecture (regex, keyword, learned classifier scoring propositions independently, or class-aware aggregator with g built from those signals) can escape the wall. **The only architectural escape is importing genuine semantic access – either from an LLM (as DR6/DR6e showed) or from a hand-authored g whose construction included direct semantic knowledge of D .**

Practically:

and the designer enumerates it fully.

labels came from semantic access to D that did not itself run through the ranker.

can be described precisely and the LLM's semantic reasoning is adequate for D 's domain.

the underlying g -construction had access to one of the above conditions; class-awareness by itself supplies nothing.

- **Regex matchers:** escape only when the realisation set is closed
- **Learned classifiers on independent features:** escape only when
- **LLM verifiers on natural-language prompts:** escape only when D
- **Class-aware aggregators over any of the above:** escape only when

5. Open questions

sound and complete on a *specific known subset* of $\mathrm{realisations}(D, C_{\{\text{test}\}})$ while explicitly admitting incompleteness on the rest? DR7 shows global soundness-completeness is impossible on open realisation sets; partial guarantees might still be well-defined.

(any classifier that fires includes p) inherits every unsound member's unsoundness. Intersection- g inherits every incomplete member's incompleteness. Neither escapes the theorem.

proposition realises D ," does it satisfy DR7's independent-semantic-access condition? Yes, in the same way DR6e's Claude verifier satisfied DR5's condition (b) – but the guarantee is only as strong as the LLM's semantic accuracy on D 's domain, and each LLM's accuracy has its

own limits.

- **Bounded-completeness** $\mathcal{G}$. Can we characterise $\mathcal{G}_D$ that are
- **Ensemble- $\mathcal{G}$** . Does combining multiple partial- $\mathcal{G}$ s help? Union- $\mathcal{G}$
- **Semantic- $\mathcal{G}$ from LLM**. If $\mathcal{G}_D$ is defined as "an LLM says this

6. Relation to the arc

DR5 stated a structural wall on proposition-ranking. DR7 shows that the wall transfers to any construction of the grouping function that DR5 proposes as an escape. Read together, DR5 and DR7 characterise the class of verification problems where no proposition-independent architecture can succeed and identify the specific structural resource (semantic access to $\mathcal{D}$) that is necessary and sufficient for escape.

DR5 + DR7 + DR5b + the six empirical papers (DCR1f, DR6, DR6c, DR6d, DR6e, and DCR2a as the class-aware companion) constitute a completed theoretical-empirical bundle. The wall's shape, empirical scope, and architectural implications are now on record. The DR-arc has finished its statement of the problem.

Appendix: reproduction

DR7 is a theorem paper; there is no code artifact. The empirical instances cited (DR6d Case B, DCR1f Case A, DCR2a Spencer collapse) each reproduce via their respective runners.